

The Taylor tree for $d = 2$ with arbitrary starting exponents

Claude, with Daniel Murfet

2026-09-07

Abstract

Completion of Astra #8 rank 3: the paper's Theorem (Taylor Tree) for $d = 2$ without any restriction on (h, k) . With $p_1 = (h_1 + 1)/k_1$, $p_2 = (h_2 + 1)/k_2$ independent, for analytic ξ, η (weighted-summable coefficient arrays at a radius $\rho > b$) and for every T ,

$$Z(N) - \sum_{\alpha \in \Lambda(h, k), \alpha < 2T} N^{-\alpha} (A_\alpha \log N + B_\alpha) = O(N^{-2T} (1 + \log N)),$$

with the canonical coefficients of unit 121 unchanged and independent of T (`twoD_taylor_tree_general`). The cutoffs are $q = \max(2T, p_1, p_2) + 1$, $M_\ell = \lceil k_\ell(q - p_\ell) \rceil$ (Astra #9); the retained pole set is $\{\alpha \in \Lambda : \alpha < 2T\}$ exactly (`mem_polesBelowGen_iff`) and every retained pole is at least $\min(p_1, p_2)$. Zero `sorry/axiom`.

1 The things formalised

```
theorem uExp_lt_vExp_general (p p : ) (h h k k M M : ) (hk : 0 < k) (hk : 0 < k)
  (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p)
  (hM : p + ((M : ) - 1) / k < p + (M : ) / k) (i j : ) (hi : i < M)
  (hj : M j) : uExp h k i < vExp h k j
theorem mergedSum_eq_canon_general ... :
  mergedSum b h h k k M M a b j c N
  = poleSet h h k k M M,
    N ^ (-) * (canonA h h k k c * Real.log N
    + canonB b h h k k a b j c )
noncomputable def cutoffQ (p p T : ) : := max (2 * T) (max p p) + 1
noncomputable def cutoffMg (k : ) (p q : ) : := (k : ) * (q - p)
theorem cutoffMg_bounds (k : ) (hk : 0 < k) (p q : ) (hq : p + 1 q) :
  p + ((cutoffMg k p q : ) - 1) / k < q q p + (cutoffMg k p q : ) / k
noncomputable def polesBelowGen (h h k k : ) (p p T : ) : Finset
theorem mem_polesBelowGen_iff ... ( : ) :
  polesBelowGen h h k k p p T
  < 2 * T (( i, uExp h k i = ) j, vExp h k j = )
theorem twoD_taylor_tree_general ( b p p : ) (h h k k : ) (h : 0 < )
  (hb : 0 < b) (hb : b < ) (hk : 0 < k) (hk : 0 < k)
  (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p)
  (x y : x → ) (hx : WSummable x) (hy : WSummable y) (T : ) :
  (fun N : => twoDAmp b N h h k k (anaAmp (ampCoeff x y) b)
  - polesBelowGen h h k k p p T,
    N ^ (-) * (canonA h h k k (fun i j s => ampCoeff x y (i, j) s)
    * Real.log N
    + canonB b h h k k (anaFaceU (ampCoeff x y) b)
    (anaFaceV (ampCoeff x y) b) (fun i j s => ampCoeff x y (i, j) s) ))
```

```

=0[atTop] fun N : => N ^ (-(2 * T)) * (1 + Real.log N)
theorem min_le_of_mem_polesBelowGen ... (h : polesBelowGen h h k k p p T) :
  min p p

```

2 Mathematics

The regrouping of units 122–123 goes through verbatim once the two ordering facts are re-proved from the shifted compatibilities: for $i < M_1 \leq j$, $\alpha_i \leq p_1 + (M_1 - 1)/k_1 < p_2 + M_2/k_2 \leq \delta_j$, and symmetrically. For the cutoffs, $q - p_\ell \geq 1$ gives $k_\ell(q - p_\ell) \leq M_\ell < k_\ell(q - p_\ell) + 1$, hence $p_\ell + (M_\ell - 1)/k_\ell < q \leq p_\ell + M_\ell/k_\ell$: both compatibilities, remainder exponent $\geq q > 2T$, and every pole $\alpha_i < 2T < q \leq p_1 + M_1/k_1$ has $i < M_1$. Unequal starts do not remove later logarithms: they occur at every collision $\alpha_i = \delta_j$.

3 Paper-facing meaning

This is `thm:TaylorTree` for $d = 2$ under the paper’s own hypotheses (analyticity of ξ, η near the closed box) and for all (h, k) , with three refinements over the paper’s statement: canonical truncation-independent coefficients, a rigorous remainder $O(n^{-T}(1 + \log n))$ for every T , and the exact pole set $\{\mu \in \Lambda(h, k) : \mu < T\}$ with $\deg P_\mu \leq 1$ and logarithms only at collisions. The leading candidate exponent is $\min(p_1, p_2)$, with a leading logarithm only when $p_1 = p_2$ (whose coefficient is $\eta(0)/(2k_1k_2) \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta s \xi(0)} ds$ in the n -normalisation), and for $p_1 < p_2$ the leading coefficient is $k_1^{-1} \sum_j \frac{b_j + \Delta}{j + \Delta} M[p_1; 0; c_{0j}]$, $\Delta = k_2(p_2 - p_1)$ (Astra #9), a genuine integral. Remaining programme (Astra #9): explicit constants and uniform families (R5), bounded linear coefficient maps (R6), Taylor-data identification (R7), expectation commuting with coefficient extraction (R9), general- d density (R8).

4 Lean notes

- The whole unit compiled at the first attempt: it is the equal-start pair of files with the compatibility consequences re-proved and the cutoffs replaced.
- `div_lt_div_iff_of_pos_right` converts $i/k < M/k$ to $i < M$ before `exact_mod_cast`.
- `le_min` (by `linarith`) (by `linarith`) is the cleanest way to bound $2T$ by a min of two explicit expressions.